PORTFOLIO · UNREAL ENGINE 5 · C++ / BLUEPRINTS

HASSAM ASLAM
UNREAL ENGINE DEVELOPER

Unreal Engine developer building combat, the Gameplay Ability System, enemy and NPC AI, multiplayer and UI in UE 5.8. Shipped 12 Fortnite islands as Zeritress.

Field Correspondent	LLM NPC · BT · NavMesh
Ashford Hollow	GAS · AI · netcode · parties
Moshpit Studios	crowd NPC AI · tech anim
Third-person action RPG	combat · boss fight · UE 5.8
GASShooter 5.8 port · UEFN	engine upgrade · shipped

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SOLO PROOF OF CONCEPT · 2026

FIELD CORRESPONDENT

UE 5.8 · C++ · LLM tool calls · Retrieval with citations · AI Perception

A reporter NPC in a newspaper building that you talk to. A language model answers from a handbook it searches and cites; the NPC proposes actions that C++ checks before a behavior tree carries them out; it notices events, remembers them, and files reports that cite what it saw. It runs offline with a scripted model, or with Claude or any OpenAI-compatible server.

THE MODEL ASKS, C++ DECIDES

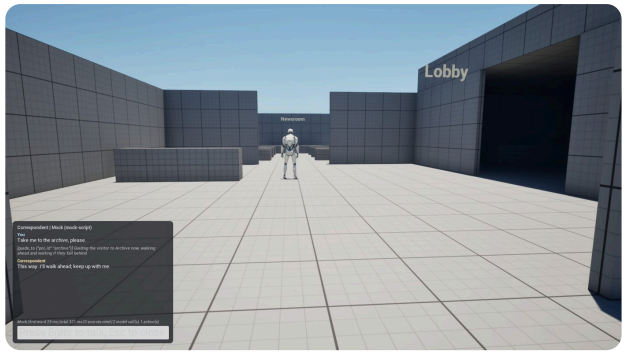
- Five tools: `guide_to`, `follow_player`, `wait_here`, `investigate`, `file_dispatch`. The model never moves the pawn.
- Every call is checked against a whitelist and a NavMesh path test; refusals go back to the model with a reason it explains to the player.
- A new message aborts the running branch: the NPC stops mid-walk in about 0.2 s.

BEHAVIOR AND MEMORY

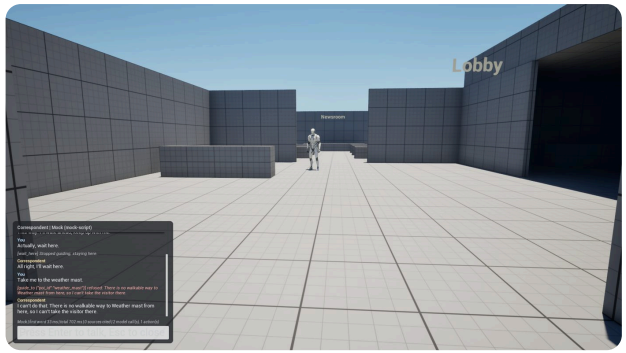
- One behavior-tree branch per mode (Investigate, Guide, Follow, Idle) with six custom C++ nodes; Detour crowd following.
- AI Perception (sight and hearing) turns events into timed memories; a dispatch is refused unless it cites one.

MEASURED

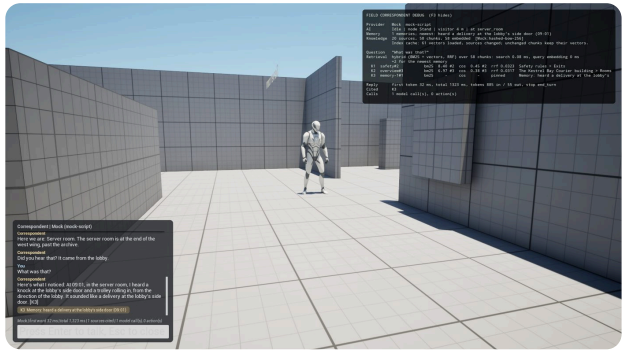
- Hybrid search (BM25 + vectors + rank fusion): 0.044 ms median over 57 chunks.
- 120 automation tests, 50 end-to-end checks passing in a packaged build.



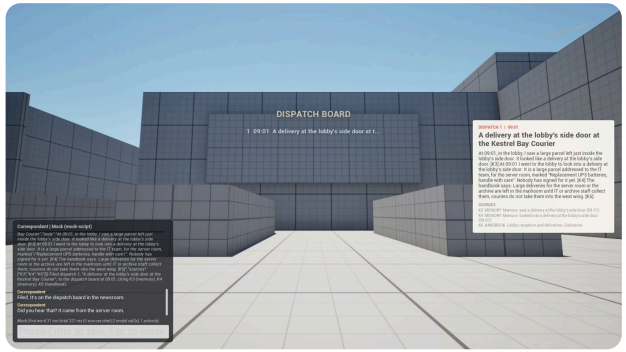
"Take me to the archive": a validated `guide_to` call; it walks ahead and waits if you fall behind.



Interrupted mid-walk, then a refused request: no walkable path to the weather mast.



Debug overlay: retrieved chunks and scores, and the answer citing a heard delivery.



A filed dispatch on the newsroom board, citing memories and the handbook.

Hassam Aslam (Zeritress)

SOLO PROJECT · 2026

ASHFORD HOLLOW

UE 5.8 | ~49,000 lines C++ | Gameplay Ability System | 2-4 players, server-authoritative

A medieval top-down pixel-art RPG slice for two to four players over a listen or dedicated server, across five streamed zones. The server decides every outcome; clients predict only what makes the game feel instant. Every map and behavior tree is generated by C++ editor commandlets.

COMBAT ON GAS

- Ability system on the PlayerState, custom attribute sets, damage and cooldown effects, gameplay tags and hit cues with predicted feedback; statuses as data.
- 3-hit melee combo with input buffering, a predicted ranged attack, and a two-phase boss whose phases are data, with telegraphs synced to server time.
- Items, crafting at stations, bombs and tools, forge upgrades and player trading, saved per player.

ENEMY AI

- One AI Controller with AI Perception; five behavior trees sharing one Blackboard; six custom C++ nodes.
- Wolf pack with flank slots and attack tokens; an archer that picks firing spots with EQS; ambushers; smoke that hides you from AI sight and bait that draws wolves off; NavMesh with Detour crowd avoidance.

NETCODE, MEASURED

CHANGE	MEASURE	BEFORE	AFTER
Server-side rewind	ghost hits, 300 ms	30%	2%
Soft collision	corrections, wolf fight	534	121
Predicted bog slow	corrections, wading	184	0



Day, rain, dusk and night: the server replicates the rule (clock anchor, weather change time), not the value. 0 B/s extra bandwidth.



The Watchtower Guardian's slam telegraph.

```
BT_Wolf
Selector Wolf  [Track Target]
 └ Hunt › Bite or Circle
    ├ Bite [Cooldown 1.4s][Token ×1]
    │  ├ Lunge [Time Limit 1.5s]
    │  ├ Use Ability: melee (GAS)
    │  └ Wait 0.4s
    ├ Circle Target (flank slot)
    └ Wait 0.2s
 └ Return Home › Move To › Idle
```

SOLO PROJECT · 2026 · CONTINUED

ASHFORD HOLLOW

AI Perception · Behavior trees · Timestamped inputs · Custom relevancy · Server-side validation

A village that keeps its own hours, a parry the server judges fairly under lag, and parties and chat.

VILLAGERS AND A WORLD THAT REACTS

- Six villagers on the enemies' AI stack: AI Perception (sight, and hearing for combat noise) and `BT_Villager`, whose schedule service turns fright, rain and the hour into where to be.
- One replicated struct per villager, its activity and start time: every client derives the work beats, lines and chatter from it. A lamplighter lights each street lamp by hand at dusk.
- Wildlife simulated on each client with no actor per animal; grass that parts around every character from one render-target buffer; footsteps by surface; props in the replicated wind.

GUARD AND PARRY

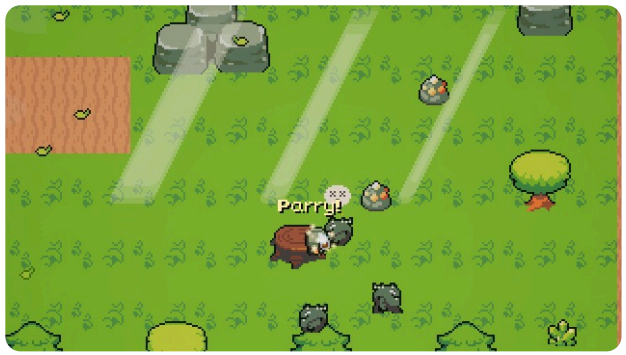
- A block in the 220° arc costs stamina instead of health; a press within 0.2 s is a parry that staggers the attacker and doubles your next hit. Unblockable attacks are violet and hatched.
- Presses carry the server time the player saw; the server bounds the claim by that player's measured lag, holds hits on them that long and judges the guard at the hit.
- **Measured:** a bot pressing at the last moment, 100 ms each way, parried **34 of 37** bites with the hold and 0 of 22 without it.

PARTIES AND CHAT

- A party replicates to its members only: an outsider's machine holds no party actor. Every request is validated and rate-limited.
- Zone, party and whisper chat: the server rate-limits, cleans and filters every line and routes it by its own facts. Zone bubbles are replicated state, so they show in replays.



Dusk: the lamplighter has lit the street, and the villagers head home talking. Every client hears the same lines.



A parry: the wolf reels; the next hit does double.



Zone and party chat: bubbles stack and never cover a name.

STUDIO WORK AND AN ACTION RPG

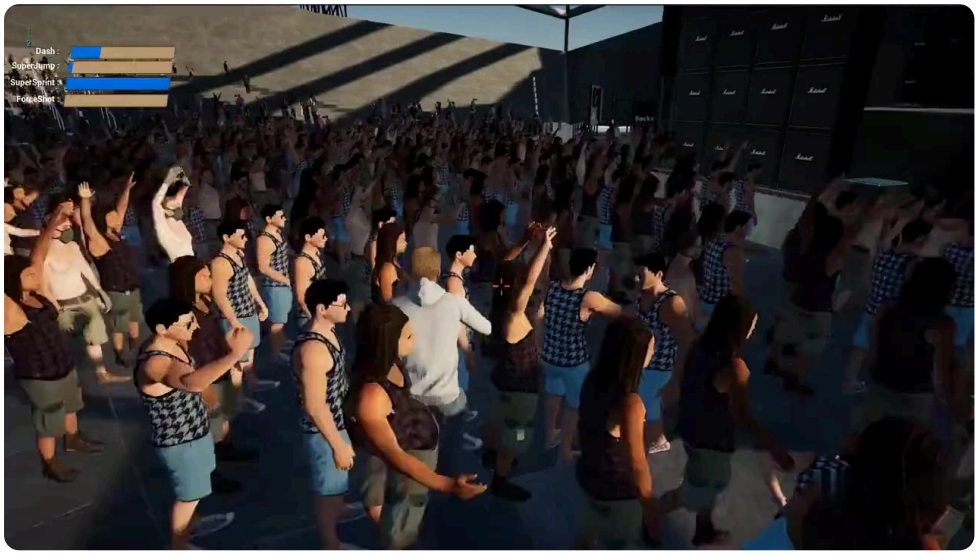
MOSHPIT · ACTION RPG

UE5 · AI Controllers · Behavior trees · Technical animation · Combat · Boss fight

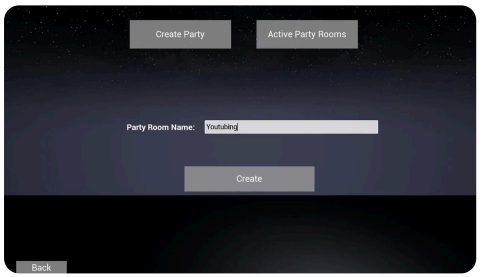
MOSHPIT VIRTUAL CONCERT · MOSHPIT STUDIOS

A multiplayer virtual concert: create or join a party room, then move through a stadium crowd of NPCs while video plays on the stage screens.

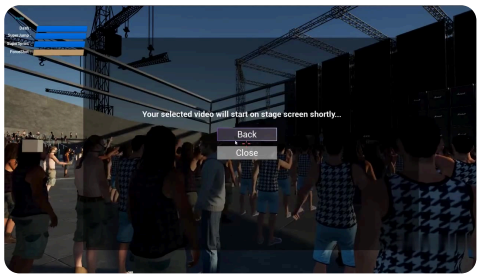
- **My part:** NPC behavior with AI Controllers and behavior trees, technical animation for crowd and player characters, and testing NPCs against player powers.
- **In the build:** Dash, Super Jump, Super Sprint and Force Shot (knocks crowd NPCs into ragdoll); party rooms; an in-game browser that puts a chosen video on the stage.



Inside the crowd.



Creating a party room.



Sending a video to the stage.

THIRD-PERSON ACTION RPG · UE5 TO UE 5.8

An action game across two levels, a bamboo shrine and a neon city, carried forward in UE 5.8 to a soulslike boss fight.

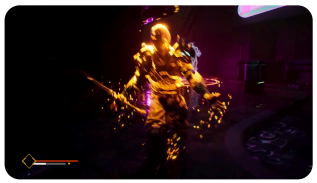
- Melee combat with ice, fire and energy-slash abilities, each with its own screen effect; enemy AI that tracks the player and attacks in groups; XP, levelling and a three-branch skill tree.
- **In UE 5.8:** a knight retargeted at runtime onto the existing combat, and the Stone Golem boss, with a souls-style boss bar whose damage trail lags, then drains.
- Inventory, crafting and loot in C++, wired to the Blueprint combat through reflection; a Dark Souls-style C++ HUD; the skill tree restyled at runtime; 10 automation tests.



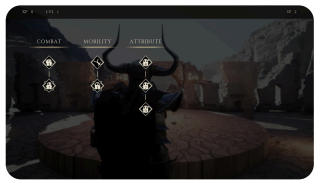
UE 5.8: the Stone Golem's bar with its flurry total, and the knight's damage trail.



Energy slash, bamboo shrine.



Fire form, neon city.



UE 5.8: the skill tree, restyled.

Gameplay reels: hassam28.github.io/unreal

Hassam Aslam (Zeritress)

ENGINE WORK AND SHIPPED TITLES

GASSHOOTER · UEFN

UE 5.3 → 5.8 | GAS weapons | Verse | Live-ops

GASSHOOTER PORTED TO UE 5.8

tranek's open-source GAS shooter sample (rifle, shotgun, rocket launcher on GAS) brought from 5.3 to 5.8, with the editor target building cleanly.

- Montage replication moved off the removed `RepAnimMontageInfo.AnimMontage` field to `.Animation` / `GetAnimMontage()`.
- Deprecated NonInstanced abilities moved to InstancedPerActor; ability tags to `Get/SetAssetTags()`; input-pressed prediction keys read from the ability instance.
- Fixed an inverted `ensure` in `GSAT_WaitTargetDataUsingActor` that always returned before registering server-side target-data callbacks.
- Documented how the weapons use GAS: abilities granted on pickup and gated by the current weapon, clip ammo replication paused while firing, predicted weapon switching, one batched RPC per shot.

FORTNITE ISLANDS IN UEFN · AS ZERITRESS, 2023 – PRESENT

12	855M+	44,962	482
islands shipped	minutes played	all-time player peak	published versions

- Six creator partners; flagship PlayStation vs Xbox at 552M minutes; Fort Lock: Soccer and Fashion Show: 50 Players.
- Verse systems: persistent economies and save state, shops and item granting, per-player state tested with 50 players, prestige and ranked progression, NPC customer AI.
- Live-ops at about two updates a week on flagship islands. Full roster at hassam28.github.io.

LET'S BUILD IT

Gameplay, combat, AI and UI programming in Unreal Engine 5. Available remotely, with overlap with US and EU hours.

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